

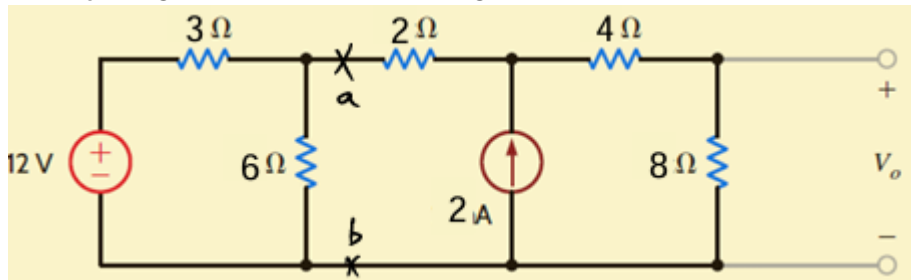
JK LakshmiPat University Jaipur

INSTITUTE OF ENGINEERING AND TECHNOLOGY

Practice sheet IV B. Tech. All, Semester I

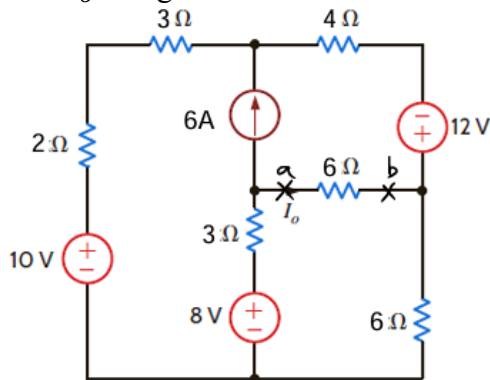
Course EE1118 Fundamental of Electrical and Electronic Engineering

1. Find V_o using Thevenin theorem in the given circuit.



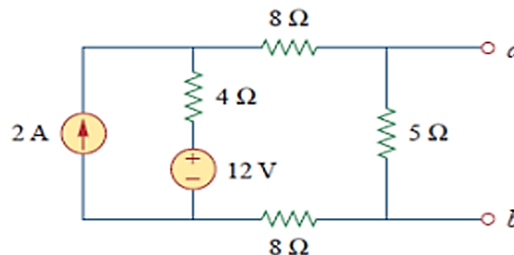
$$V_o = 8 \text{ volt}$$

2. Find I_o using Thevenin's or Norton's theorem.

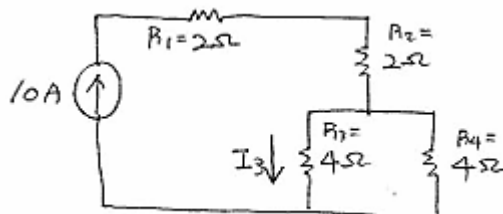


$$I_o = -2.44 \text{ Amp}$$

3. Find the Norton equivalent circuit of the circuit shown to the left of the terminals a- b.

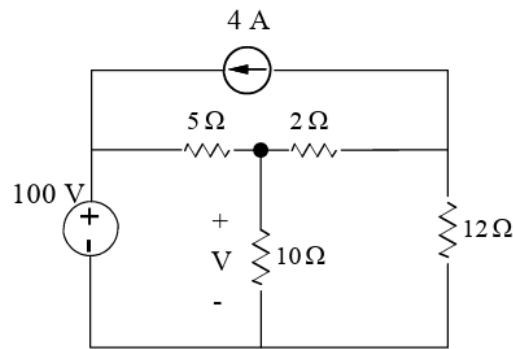


4. Find the current I_3 of the circuit using Thevenin theorem.



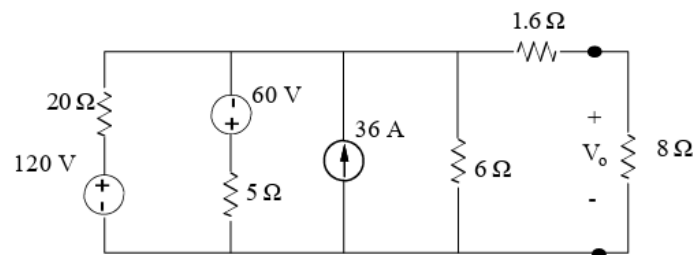
$$I_3 = 5 \text{ Amp}$$

5. Use Thevenin Theorem to find V.



$$V = 50 \text{ volt}$$

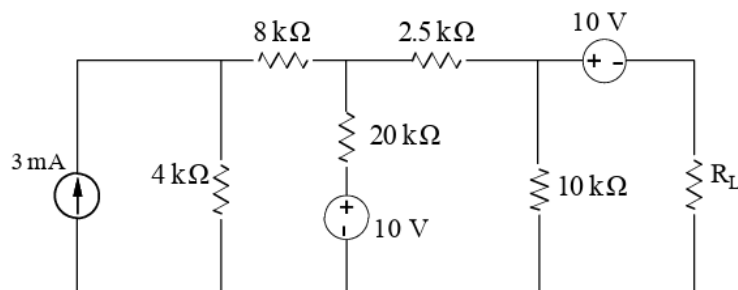
6. Find V_O using Thevenin's or Norton's theorem.



$$V_O = 48 \text{ volt}$$

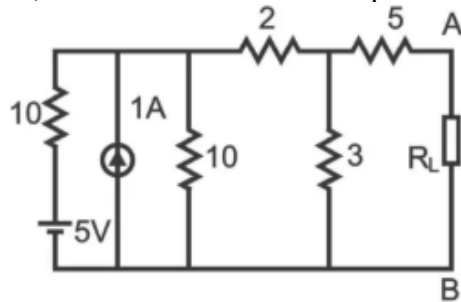
7. For the given circuit find:

- R_L for maximum Power Transfer
- Maximum Power transfer to R_L



$$R_L = 5 \text{ k}\Omega, \quad P_{L \max} = 0.957 \text{ mW}$$

8. In the given circuit, obtain the condition from maximum power transfer to the load R_L . Hence, determine the maximum power transferred.



$$R_{th} = 7.1 \text{ ohm} \quad \text{and} \quad P_{max} = 0.178 \text{ W}$$